

TUTORIAL – 3: Requirement Reviews & Inspections

Aim

To understand **Requirement Reviews, Code Walkthroughs, Inspections, and Peer Review activities** using sample project documents and identify defects before software execution through static testing.

Description

Static Testing is a software testing technique used to identify defects **without executing the program**. It helps improve software quality in the early stages of the Software Development Life Cycle (SDLC).

The main static testing techniques are:

- **Requirement Review:** Reviewing the Software Requirement Specification (SRS) to identify missing, incorrect, or ambiguous requirements.
- **Code Walkthrough:** An informal review where the developer explains the source code to team members to identify logical errors and improve understanding.
- **Inspection:** A formal review process conducted by trained reviewers following a predefined checklist to identify defects.
- **Peer Review:** Team members review each other's work to improve quality and reduce errors before testing.

These review activities reduce development cost by detecting defects early.

Problem Statement

Perform Requirement Review, Code Walkthrough, Inspection, and Peer Review for the **Login Module of a Student Management System**. Identify defects in the requirement document and source code before execution, prepare review observations, and document the testing activities.

Implementation Details

The Student Management System provides secure login functionality for authorized users.

The following review activities are performed:

1. Review the SRS document for completeness.
2. Perform a Code Walkthrough of the Login Module.
3. Conduct a Formal Inspection using a checklist.
4. Perform Peer Review and record observations.
5. Prepare test cases based on reviewed requirements.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	Visual Studio Code / PyCharm
Input/Test Data	Username: admin Password: admin123

Source Code

Login Module

```
username = input("Enter Username: ")  
password = input("Enter Password: ")
```

```
if username == "admin" and password == "admin123":  
    print("Login Successful")  
else:  
    print("Invalid Username or Password")
```

Test Case Detailed

Field Details	Description
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	User Login Authentication
Test Scenario	Verify login with valid credentials

Test Objective	Ensure only authorized users can log in
Test Type	System Testing
Test Technique	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application is installed and login page is displayed
Postconditions	User successfully enters the Home Page
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome / Windows 10
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful
Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open Application	—	Login page displayed	Login page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass

3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login Button	Click	Login Successful	Login Successful	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Review	Review SRS document for completeness and correctness	Review Report
2	System Design	Inspection	Perform formal inspection using checklist	Inspection Report
3	Development	Code Walkthrough	Review source code with development team	Walkthrough Report
4	Development	Peer Review	Team members review each other's code	Peer Review Report
5	Testing	Test Case Preparation	Prepare test cases from reviewed requirements	Test Case Document

Observation

Sr. No.	Observation	Remarks
1	All login requirements are clearly defined	Pass
2	Source code follows required login logic	Pass
3	No ambiguity found in SRS document	Pass
4	Peer Review suggested improving code comments	Improvement Suggested

5	Inspection checklist completed successfully	Pass
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Output

Requirement Review Completed Successfully

- ✓ SRS Reviewed
- ✓ Code Walkthrough Completed
- ✓ Inspection Completed
- ✓ Peer Review Completed

Login Test Result:

Username : admin

Password : admin123

Output:

Login Successful

Solve the Below Mention Questionnaires

1. Which testing technique identifies defects without executing the software?

A) Dynamic Testing

B) Static Testing

C) Regression Testing

D) Performance Testing

Answer: B) Static Testing

2. Which document is primarily reviewed during Requirement Review?

A) Source Code

B) Database Schema

C) Software Requirement Specification (SRS)

D) Test Report

Answer: C) Software Requirement Specification (SRS)

3. What is the primary purpose of a Code Walkthrough?

A) Execute software

B) Deploy software

C) Review code and identify defects

D) Create database tables

Answer: C) Review code and identify defects

4. Inspection is a _____ review process.

- A) Informal
- B) Dynamic
- C) Automated
- D) Formal**

Answer: D) Formal

5. Peer Review is performed by:

- A) Customers
- B) End Users
- C) Team Members Reviewing Each Other's Work**
- D) System Administrators

Answer: C) Team Members Reviewing Each Other's Work

Conclusion / Outcome

- Understood the concepts of **Static Testing** and software review techniques.
- Performed **Requirement Review** on the Software Requirement Specification (SRS).
- Conducted **Code Walkthrough**, **Formal Inspection**, and **Peer Review** for the Login Module.
- Identified potential defects before software execution, improving software quality and reducing development cost.
- Learned how review activities support the SDLC by detecting defects early and preparing effective test cases.